



Logic Realization of Galois Field for AES SBOX using Quantum Dot Cellular Automata

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Abstract

With the growing technological trends in VLSI domain, quantum dot cellular automata (QCA) technology is slowly replacing CMOS technology due to its smaller feature size, high operating frequency and reduced power consumption. In the initial research phase, QCA has been used to implement various combinatorial and sequential circuits models, which are the fundamental blocks in various applications. Nowadays, researchers focus on the implementation of application-based designs using QCA. This motivated to implement the Galois field (GF) functions for SBox module in the most secure cryptography encryption standard AES with QCA. In AES, SBOX is the predominant power consumption modules. Hence, a research has been carried out to implement a compact QCA-based AES-SBOX with GF. This paper describes the implementation of our proposed QCA-based AES-SBOX with Galois field and analysis of various GF functional modules in terms of area, performance, energy and QCA cells used. The functional verification is performed using the simulated waveforms.

Keywords QCA · Galois Field (GF) · AES SBOX · Cryptography Algorithm · QCA Designer

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1 Introduction

In the semiconductor industry, microelectronics has made remarkable achievements over the past four decades based on the Garden Moore's law. Minimizing size, fast switching speeds and low power consumption are open challenges for VLSI's CMOS technology to keep Moore's law in effect. QCA is a possible alternative to CMOS technology and promises a successful digital architecture with high system density and low power loss in the future. It replaces today's CMOS technology where the CMOS technology reaches its fundamental limits [1–3]. Cryptography plays a key role in secure data transmission in the information and communication field. Depending upon the input format, substitution may have a single bit or multiple bits. Substitution plays a significant key role in most of the cipher models, which may be simple substitution or complex transforms-based substitution. The implementation may be either a lookup table or Boolean function-based logic. Perhaps, the most secure cryptographic algorithm DES was broken by the DES-Challenge DESCHAL [4]. In 2000, Rijndael Advanced Encryption Standard (AES) was approved as the best secure algorithm that replaces DES. Its operation is primarily based on the substitution function. An AES encryption standard is one of the best encryption algorithms in the security field. This paper describes the design of basic functional blocks in AES using QCA. The size of QCA circuit depends on the total number of cells used in it. Also, area usage, cell area, total area and quantum cost are dependent on circuit complexity. Hence, utmost care of these parameters is taken in this work and performance is analyzed.

The paper is organized as given. Section 2 briefs the related work. The background theory about AES and QCA is explained in Sect. 3. Section 4 details the QCA implementation of AES S-Box GF functions. The simulation results and discussions are described in Sect. 5. The conclusion and future scope are given in Sect. 6.

2 Related Work

AES has been implemented using Coprocessor, FPGA, ASIC and SOC with major objectives of achieving minimized power consumption, reduced area and increased operating speed. QCA is one such design style used in the nanodesign approach. In this literature review, we examined various QCA design implementations in three broad categories: basic logic gate design, combinatorial and sequential circuit design, and application-based design. Numerous QCA implementations have so far been carried out by researchers in various fields, such as the basic design of logic gates in [5] and energy-efficient XOR in [6]. Furthermore, few designs for full adder cells have been reported in [7, 8]. The energy dissipation analysis for simple multiplexer is done by [10] and demultiplexer by [11]. In [12], authors discussed the ALU circuit using reversible techniques to

minimize the area. A 8-bit adder and subtractor are designed by [13], and efficient reversible adder had been proposed [9]. Recently, the research focused on the QCA-based design with energy estimation analysis. In [14], authors had discussed the QCA implementation of the AES finalists Serpent Block Cipher SBOX module, including the complexity, area and delay of the 4-bit SBOX which is based on logic model and Lookup Table model(LUT). The detailed implementation of SBOX using the LUT model was suggested in [15]. In [37], the authors proposed the QCA-based Serpent SBOX implementation for crypt analysis attacks. They proposed the differential power analysis attack and analyzed the attack by using different key modules. A simple XOR-based cryptography model was proposed in [38], where plain text and keys were given as input to the XOR module, which acted as the encryption module. Similarly, output called cipher text is given to input to another XOR. This was treated as a decryption module. A 5/1 stream cipher was implemented by [39] for the GSM network. In this QCA implementation, memory cell, majority function and shift register's property were analyzed. A simple nano-router and cryptography architecture model were proposed in [40] with help of basic 4:1 mux and 1:4 demux. They analyzed the power dissipation with different tunneling energy as well as the design complexity of the proposed design. An image masking-based encryption was proposed by [41], and the encoder and decoder model was given. Also, they analyzed different crypt analysis methods with different tunneling energy models. A novel encryption model of Select Cross and Variation (SCV) was proposed by Kundu et al. [42], in which they proposed the reversible SCV gate for the encryption process. Based on the above literature work, it is found that there is a huge gap in research analysis of advanced encryption standards leading to the requirement of efficient AES and also their implementation using QCA.

3 Background

3.1 QCA–Quantum Dot Cellular Automata

QCA was first introduced by Lent [16]. Today, this QCA nanostructure offers a new concept for encoding, data transmission and processing of binary information. An array of tiny nano-structured quantum dots is arranged for the computation of each operation. The logic states of QCA cells are dependent on the position of the electrons rather than the voltage cell.

QCA cells are classified into two types, namely type 1 model and type 2 model based on the location of four points in the QCA structure. In type 1, as shown in Fig. 1 a, four points are placed in the corner of the cell, while in type 2, the four points are in the center sides of the cell. These type 1 model and type 2 model cells are shown in Fig. 1. In general, we use the type 1 QCA structure. The individual electrons communicate with each other through the use of Colombian repulsion in the QCA architecture, which replaces the existing current switching transistor. The operation of the QCA circuit depends on the interaction between cells. In addition, the cell information is arranged in the polarization of the electrical charges in such

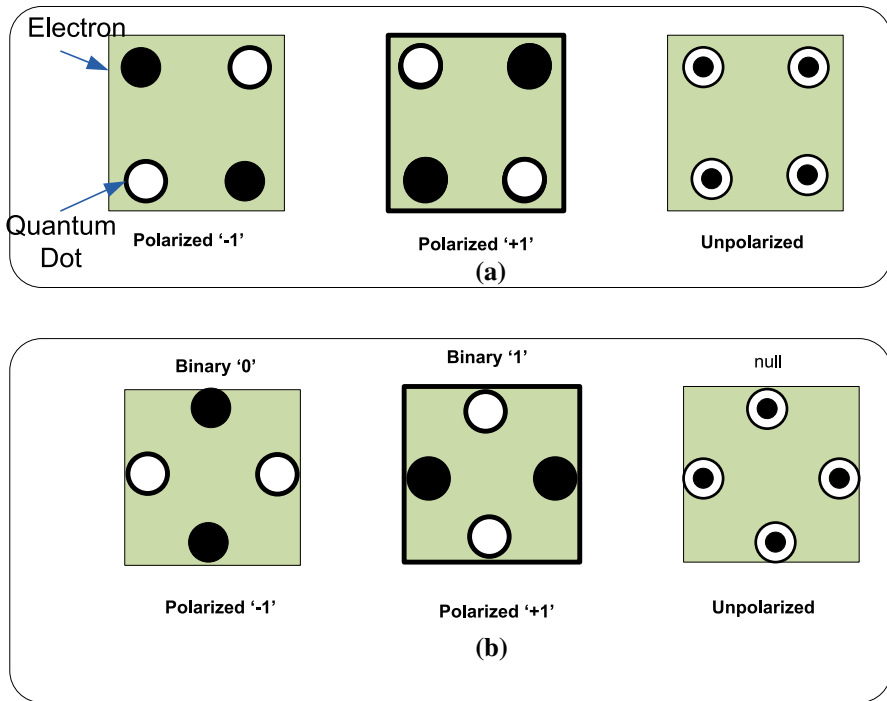


Fig. 1 **a** Type 1 model QCA cell, **b** type 2 model QCA cell

a way that charges do not transport. During operation, the neighboring cell creates a dynamic state and settles into one of the possible states [3, 16, 17]. In QCA, electrons are not permitted to jump between cells, whereas within cells, they are permitted to jump. This phenomenon is attained through quantum mechanical tunneling. The barrier between the cells is sufficient to effectively prevent the inter cellular tunneling effect. Because of the Columbic interaction between the charges, the electron is only in the diagonal antipodal places, which can be displayed when numbering clockwise. A polarization P can be measured by electric charges which is depicted by the following equation [18].

$$P = \frac{(\rho_1 + \rho_3) - (\rho_2 + \rho_4)}{\rho_1 + \rho_2 + \rho_3 + \rho_4} \quad (1)$$

where $\rho_1, \rho_2, \rho_3, \rho_4$ are the electric charges. The polarization of the charges in the cells consists of only two probable states either one or zero, i.e., $P=+1$; $P=-1$.

3.1.1 QCA Gates

Similar to digital circuit design, the gates, interconnection and clock signals also play a key role in the QCA.

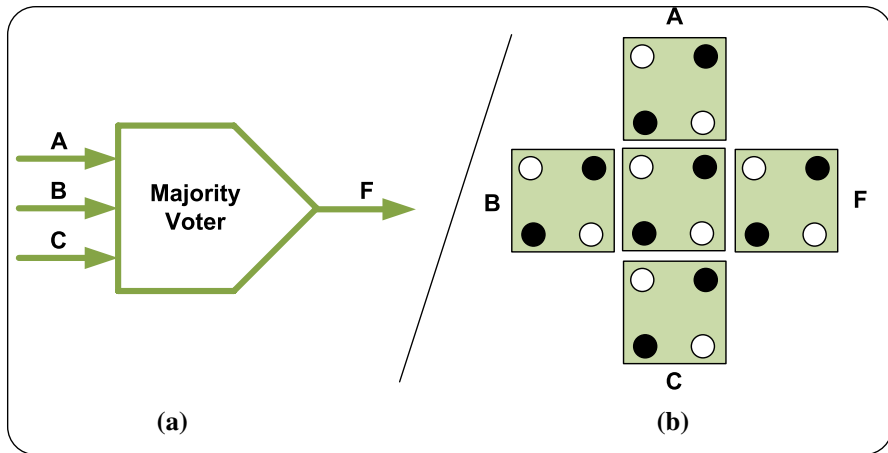


Fig. 2 **a** Majority Gate or Majority Voter; **b** QCA Majority Gate

A majority gate (MG) is a basic logic element of QCA which is shown in Fig. 2 and is also referred to as a majority voter (MV). Here, inputs are noted by A, B, C and output is F. The output F is polarized according to the polarization of the majority of the input cells. For example, if inputs A and B are polarized to +1, and input C is polarized to -1, then output F is polarized as +1 because the majority of the input polarization is +1 [1, 16]. The three-input MG is defined by Eq. (2).

$$MG(A, B, C) = F = AB + BC + CA; \quad (2)$$

This logic can be extended to any number of inputs. The inverter is another basic element whose output polarization is opposite to the input polarization. The basic QCA inverter is shown in Fig. 3. There are two types of inverters used in QCA implementation based on circuit complexity and signal transmission. In a simple model, two-cell inverter includes two cells only based on the opposing electrons in

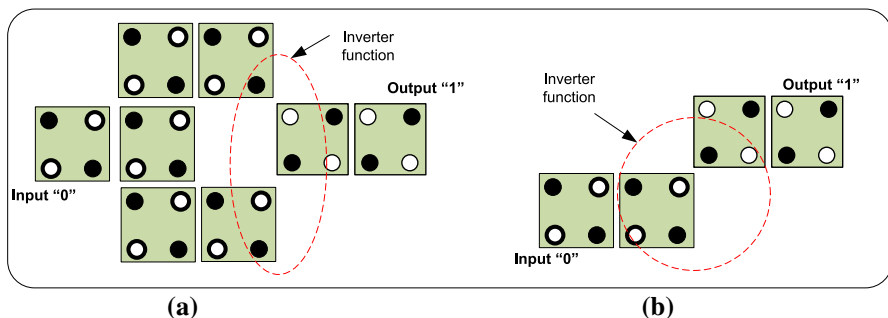


Fig. 3 Different types of inverter **a** robust inverter, **b** simple displaced inverter

the opposite corner. However, it suffers from signal integrity, and hence, another robustness type inverter is selected even though it takes more cells [2].

3.1.2 XOR gate

Apart from the basic AND, OR, Inverter gate, XOR gate is widely used in many digital circuits like parity checkers/detectors, ALU, code converters, reversible logic circuits, etc. In our design, XOR gate plays a significant role and some of the QCA XOR models are highlighted. The low complexity conventional XOR gate is introduced by different authors [19, 20]. The simple 3 input and 2 input XOR single layer XOR gate proposed by [6] is used to implement our proposed GF functions (Fig. 4).

3.1.3 QCA wires

By placing QCAs next to each other, the QCA wires are constructed. Figure 5 shows the type 1 QCA wire. When the input is placed in the left-most cell, the data are transferred to the right-most cell. The simplest practical cell arrangement of QCA wires is shown in Fig. 5. A type 1 is a simple model simple binary arrangement wire which is illustrated in Fig. 5a, whereas in type 2 5b an inverter chain has been created by placing side by side 45 degrees inverted QCA cell.

In addition, wire crossing in the QCA design is an inevitable step that is either coplanar or multi-layered [16–18].

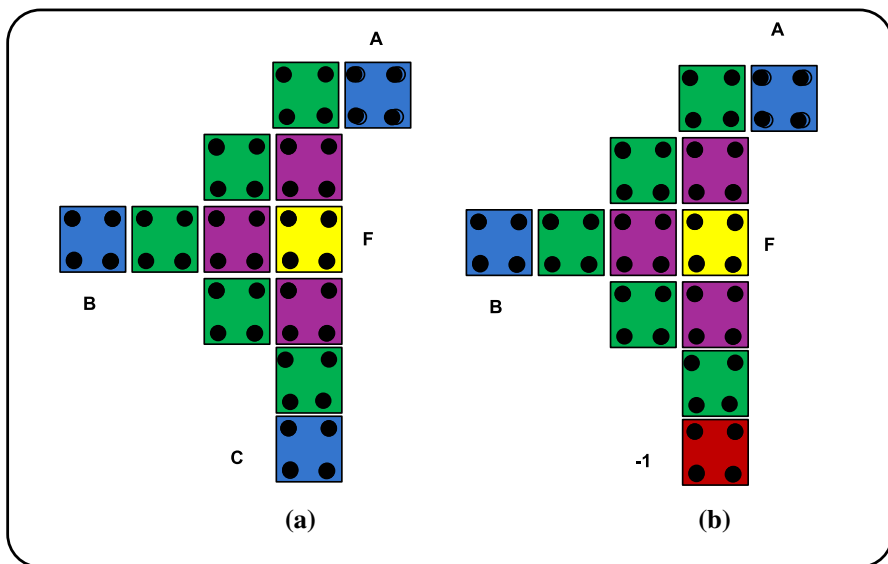


Fig. 4 3 input and 2 input XOR gate

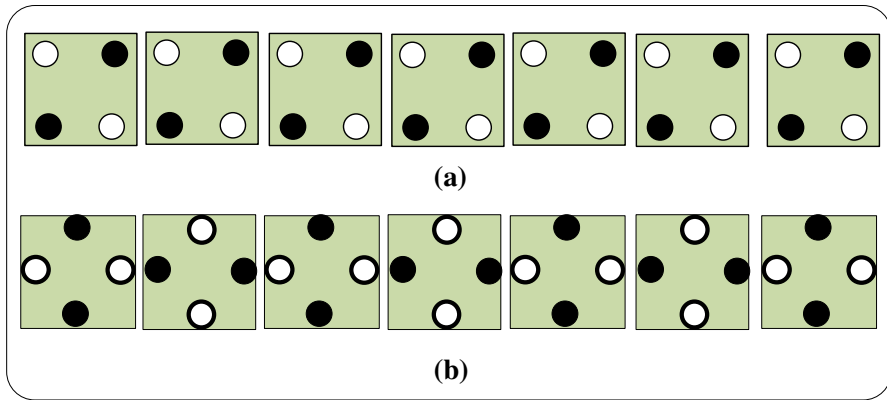


Fig. 5 a Type 1 model QCA wire; b Type 2 model QCA wire

3.1.4 QCA clocking

Similar to the clock-based CMOS circuits, inter-dot barrier of QCA is controlled by the clock signal. Any combinational or sequential-based QCA design is controlled by the pipeline-based timing mechanism. This mechanism controls the movement

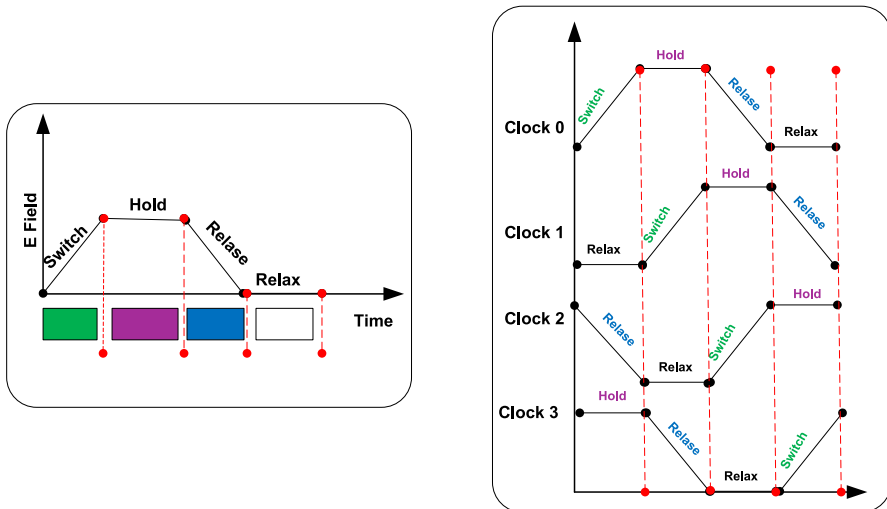


Fig. 6 Four phases QCA Clock

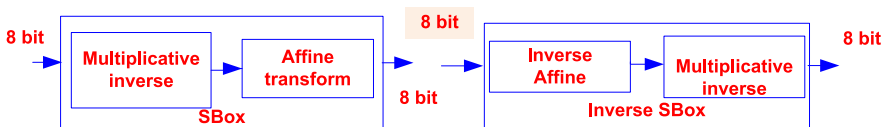


Fig. 7 AES SBOX-block diagram approach

of data and supplies the cell with power. This pipeline-based clock mechanism is divided into the four-phase model, namely Switch, Hold, Release and Relax, [16, 18]). In QCA, we adopt four clocks-CLOCK 0, CLOCK1, CLOCK 2 and CLOCK3 which are 90 degree out of phase (Figs. 6 and 7).

There are different phases of operation: In the first phase of switch, QCA cell starts the polarization and inter-dot barriers are gradually increased and QCA cell finally reaches any one of the polarizations based on its neighboring cell. In the second phase of Hold, cell has a high interdot barrier as a fixed polarization to drive the next cell; then, in release and relax phase, the cells will be in un-polarized level. These two later phases cannot be influenced to ascertain the state of the QCA cell [17, 18, 21, 22].

3.2 Advanced Encryption Standard

Rijndael AES is the final AES symmetric block cipher cryptography algorithm. This cryptographic algorithm uses the private key, 128-bit block module and different key lengths 128,192,256 depending on the number of round operations 10, 12 or 14 [23]. It consists of Add Round Key, SubByte Transformation, MixColumn Transformation and Shift Row for encryption, and Add Round Key, Inverse Sub Byte Transformation, Inverse MixColumn and Inverse Shift Row for decryption.[23, 24]. According to the NIST statement, AES is one of the best encryption standards that adopts the block cipher model. Due to their nonlinear transformation and matrix multiplication, the SBOX and Mixcolumn consume more power compared to other modules in AES. These operations imply the diffusive and confusing properties of AES. First, to perform the transform operation, input (8 bit) is mapped to another input (8 bit) data, which is based on Affine transform and multiplicative inverse function. The input 1×4 input matrix is multiplied by 4×4 matrix distance multipliers 01, 01, 02 and 03 in Mixcolumn, and 09, 0B, 0D and 0E in inverse Mixcolumn. Further, the shift row operation is performed row by row by changing the position in each byte in each row of Mixcolumn, and then, round keys are added using XOR operation [25, 26, 34].

4 QCA Implementation

As mentioned in [23], Rijndael SBOX has a less correlation between input and output. Due to this, it cannot be described by a simple mathematical expression. In addition, this low correlation property offers strong resistance to cryptanalysis attacks. To design an efficient SBOX, the designer should minimize the largest non-trivial value in XOR table and the complexity of algebraic expression in $GF(2^8)$ [27, 28]. The operation of SBOX is divided into two major functions, namely multiplicative inverse and Affine transform. In inverse SBox, inverse affine and multiplicative inverse operations are done. But, multiplicative inverse is common to both SBOX and inverse SBOX. According to [29], the AES SBOX data are represented as 8-bit input data, which are represented as an arbitrary

polynomial $bx + c$. In this polynomial, most significant nibble is referred as b and the least significant nibble is referred as c . Now, the multiplicative inverse is written in Eq. (3) [35].

As mentioned in [35], S box operation is based on one to one mapping of multiplicative inverse. A Rijndael 8-bit SBOX is represented by GF(256) finite field. This one-to-one inverse mapping which can be done in several ways. In [35], a GF(16)-based operation was introduced. A given SBOX is represented by irreducible polynomial $x^2 + Ax + B$ of degree 2, and inverse is represented as $(bx + c)^{-1}$

$$(bx + c)^{-1} = b(b^2B + bcA + c^2)^{-1} + (c + bA)(b^2B + bcA + c^2)^{-1} \quad (3)$$

where A and B are the coefficients of the irreducible polynomial $x^2 + Ax + B$. Moreover, we can have a freedom to select A and B value for Eq. 3 [35] simplified by substituting $A=1, B=\lambda$

$$(bx + c)^{-1} = b(b^2\lambda + bc + c^2)^{-1} + (c + b)(b^2\lambda + bc + c^2)^{-1} \quad (4)$$

From Eq. 4, it can be observed that the multiplicative inverse module consists of GF multiplication, GF squarer and GF lambda multiplier. Before applying the above operation, the input data should go through the isomorphic operation. The block diagram of 8-bit multiplicative inverse is shown in Fig. 8.

Symbol	Function representation
δ	Composite field isomorphic mapping
x^2	Square
$x\lambda$	Constant λ Multiplication
$\oplus$	XOR operation
x^{-1}	Multiplicative inversion;
x	Multiplication

In Fig. 8, GF(2⁴) arithmetic operations are used and δ^{-1} represents inverse mapping to GF(2⁸).

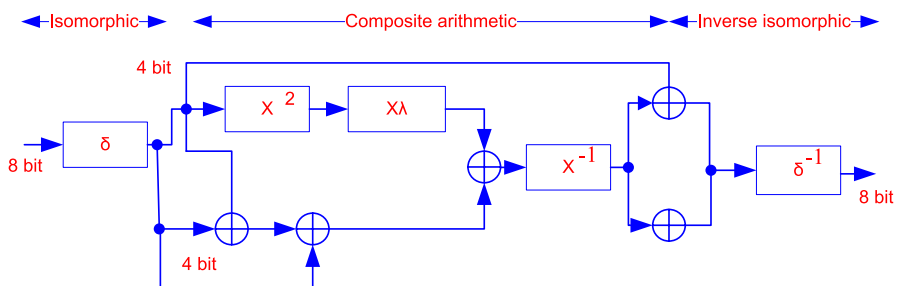


Fig. 8 8-bit Multiplicative inverse

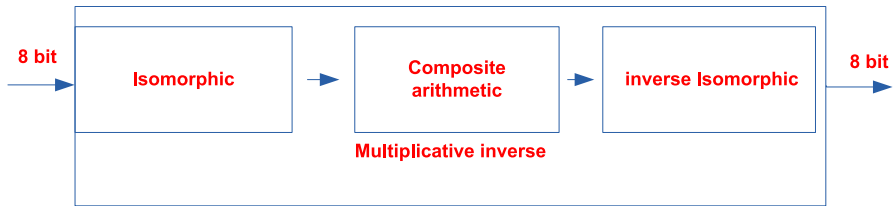


Fig. 9 Multiplicative inverse

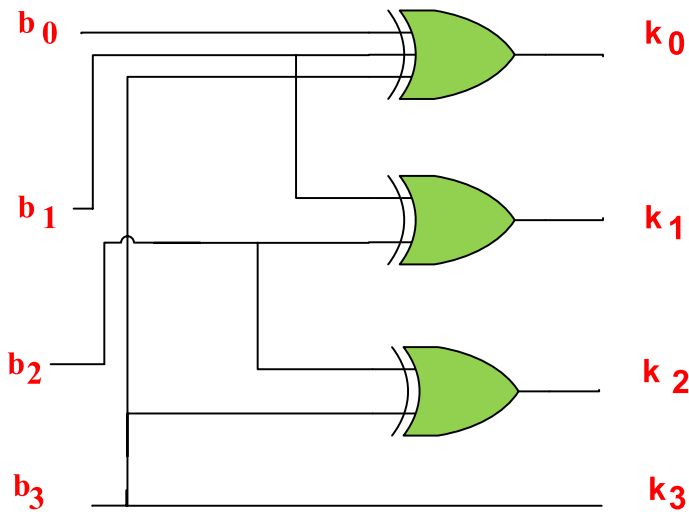


Fig. 10 Logic diagram of GF squarer

4.1 Multiplicative Inverse

The first phase involves the isomorphic operation that maps the one-to-one mapping procedure because the multiplicative inverse cannot be applied directly to the GF as it requires more multiplication operations. Therefore, an isomorphic transform is used to convert the GF into a composite arithmetic field to do the simple multiplication. Isomorph uses the bijective property to represent the data. Figure 9 shows the block diagram of Multiplicative inverse.

4.1.1 4-Bit GF Squarer

To square $GF(2^4)$, the input $GF(2^4)$ is represented in 4-bit form $[b_3b_2b_1b_0]$ and squared using GF multiplication. To perform the square operation, $GF(2^4)$ is decomposed into a lower-order field $GF(2^2)$. The resulting square value is represented as “k” which is also in binary format. To fit the resulting square values

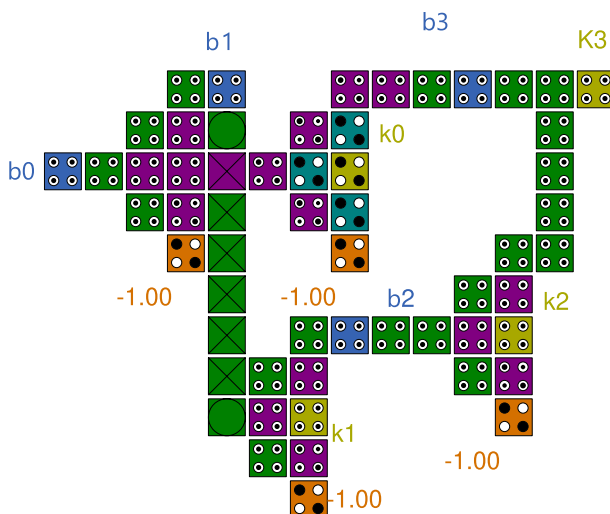


Fig. 11 QCA implementation of GF Square

into the same space, the irreducible polynomial $x^2 + x + 1$ is applied by setting $x^2 = x + 1$. The output of GF squarer is given by Eq. 5

$$[k_3 k_2 k_1 k_0] = [b_3, b_3 \oplus b_2, b_2 \oplus b_1, b_3 \oplus b_1 \oplus b_0] \quad (5)$$

Figures 10 and 11 show the logic diagram of GF squarer and QCA implementation of GF squarer, respectively.

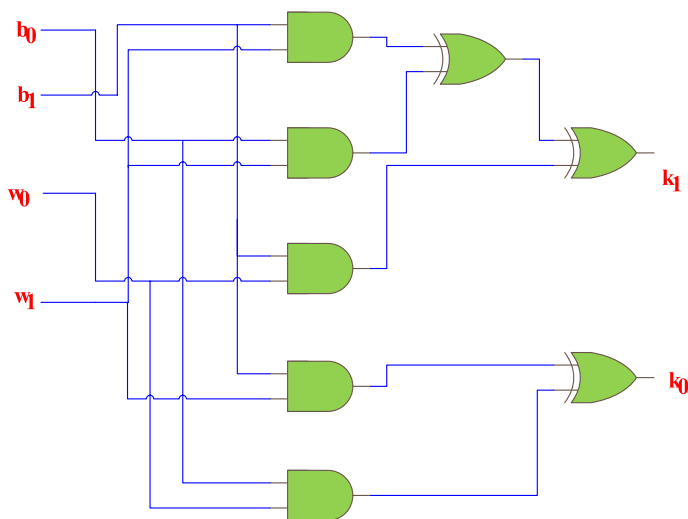


Fig. 12 Logic diagram of GF(2^2) multiplication

4.1.2 GF Multiplication

GF(2^2) multiplication is defined as $k=bw$. The terms k , b and w are represented by 2-bit binary. The simplified multiplication is represented in Eq. 6.

$$[k_1, k_0] = [b_1w_1 \oplus b_0w_1 \oplus b_1w_0, b_1w_1 \oplus b_0w_0] \quad (6)$$

Figure 12 shows the logic diagram of

Figure 12 shows the logic diagram of GF(2^2) multiplication, and Fig. 13 shows the QCA implementation of GF(2^2)

4.1.3 λ Multiplication

In GF, the lambda multiplication is defined as $k = b\lambda$ [30], where input b is multiplied by the constant 1100_2 . The terms are decomposed into binary data, which are defined in [26]. This is represented in Eq. 7.

$$[k_0k_1k_2k_3] = [b_2, b_3, b_3 \oplus b_2 \oplus b_1 \oplus b_0, b_2 \oplus b_1] \quad (7)$$

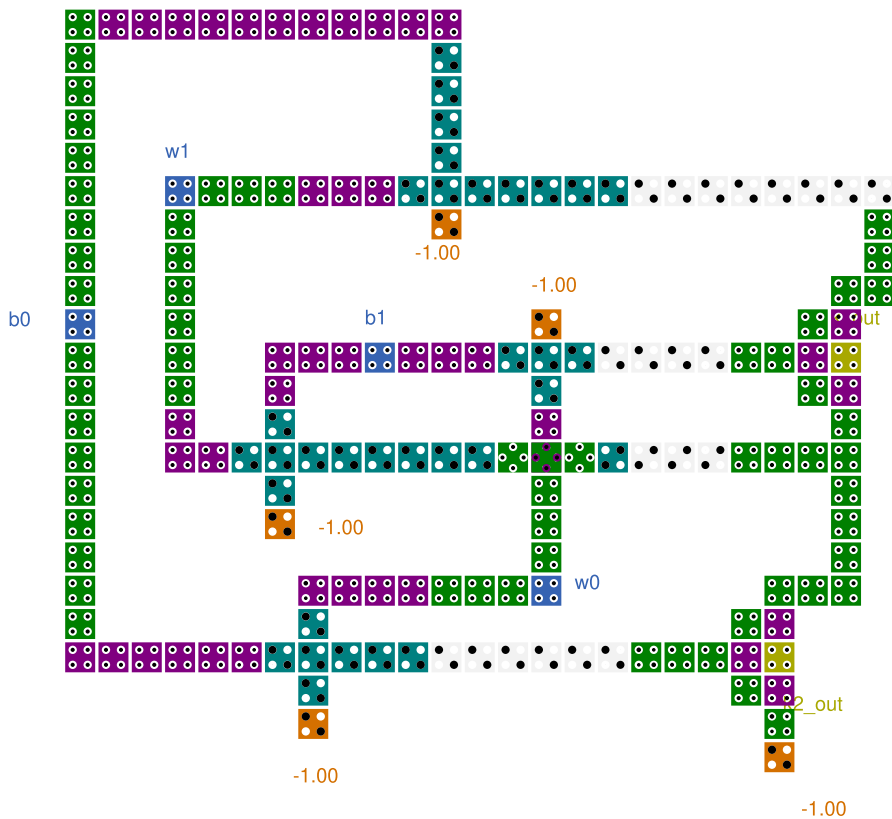


Fig. 13 QCA implementation of GF multiplication

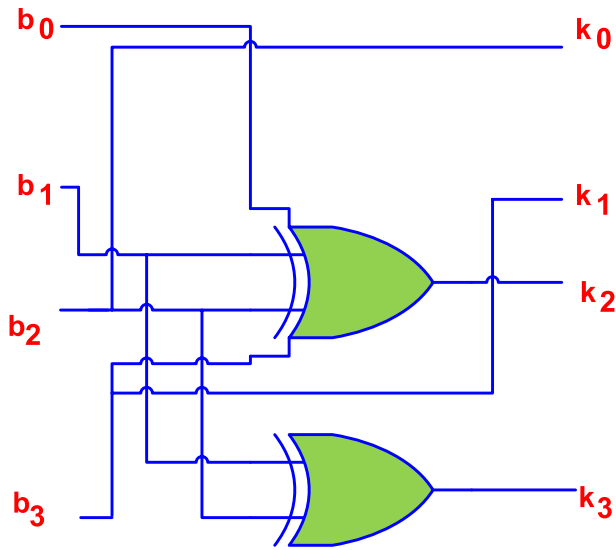


Fig. 14 Lambda λ multiplication

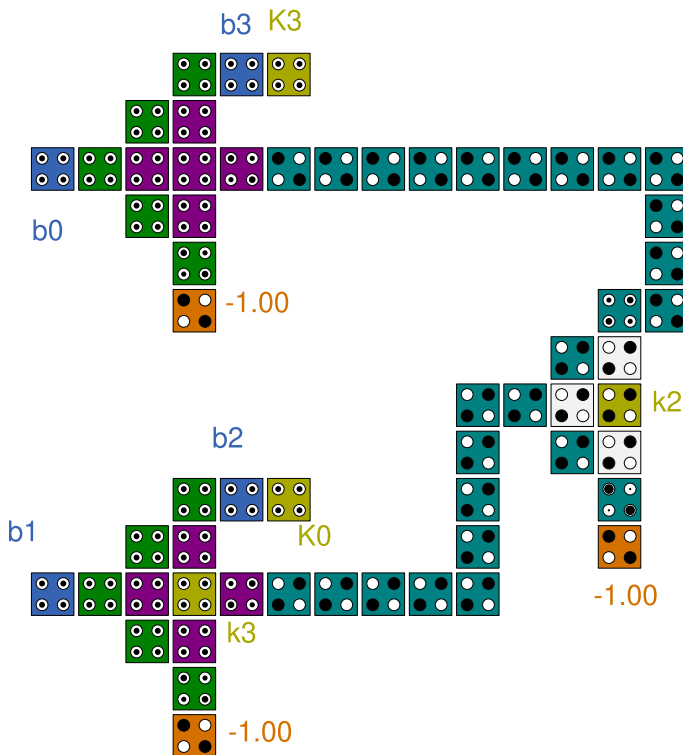


Fig. 15 Lambda λ multiplication

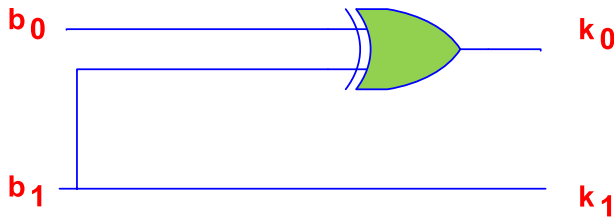
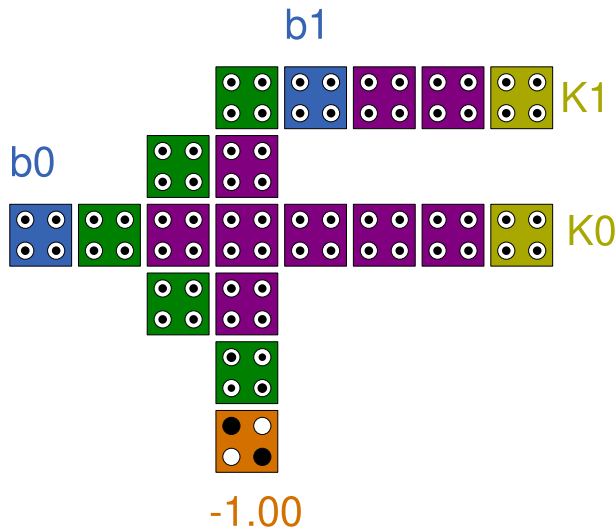

 Fig. 16 GF ϕ multiplication

 Fig. 17 QCA pi ϕ multiplication

Figure 14 shows the logic diagram of GF λ multiplication, and Fig. 15 shows the QCA implementation of GF λ multiplication.

4.1.4 ϕ Multiplication

Here the multiplication result $k = b\phi$ where $\phi = 10_2$ and input $b = [b_0, b_1]$ Eq. 8. The two-bit result is computed using Eq. 8

$$k = [k_1 k_0] = [b_1 \oplus b_0, b_1] \quad (8)$$

Figure 16 shows the logic diagram of ϕ multiplication, and Fig. 17 shows the QCA implementation of ϕ multiplication.

4.2 Isomorphic Function

Let the 8-bit input be represented as $[b_7 b_6 b_5 b_4 b_3 b_2 b_1 b_0]$ and 8-bit output be $[b'_7 b'_6 b'_5 b'_4 b'_3 b'_2 b'_1 b'_0]$.

$$\delta \times b = \begin{pmatrix} ism_1 \\ ism_2 \oplus ism_3 \oplus b_4 \oplus b_3 \\ ism_2 \oplus b_3 \oplus b_2 \\ ism_1 \oplus b_3 \oplus ism_3 \\ ism_2 \oplus ism_3 \\ b_7 \oplus b_4 \oplus b_3 \oplus ism_3 \\ ism_4 \oplus b_3 \\ ism_4 \oplus b_0 \end{pmatrix} \quad (9)$$

where $ism_1 = b_7 \oplus b_5$; $ism_2 = b_7 \oplus b_6$; $ism_3 = b_2 \oplus b_1$; $ism_4 = b_6 \oplus b_4$ represent the intermediate results for isomorphism whose logic diagram is shown in Fig. 18. The proposed logic diagram of isomorphism is shown in Fig. 19, and its QCA implementation is shown in Fig. 20.

5 Results and Discussion

5.1 QCA Simulation Setup

In this section, the detailed outcome of area, power, and delay of various functional modules, which are core segment of the AES SBOX, is presented. QCADesigner 2.0.3 simulation tool has been used for layout design and functional verifications of the circuit [31]. Another tool QCADesigner-E is used for energy estimation [32]. QCADesigner v2.0.3 has different simulation engines, namely bi-stable approximation engine and coherence vector engine. In bi-stable

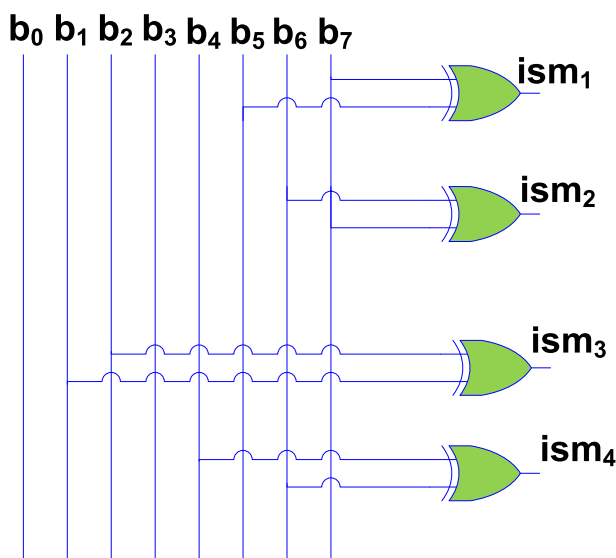


Fig. 18 Intermediate results for isomorphism

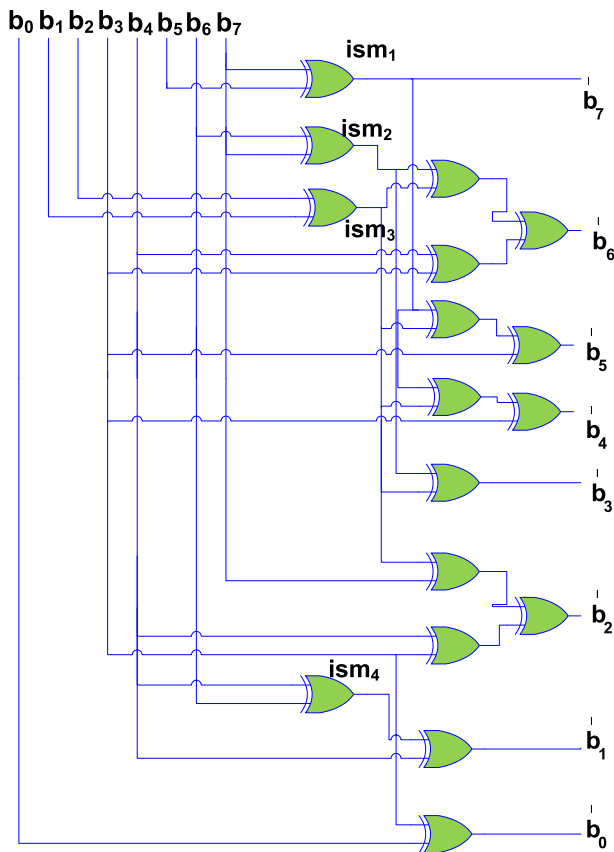


Fig. 19 Proposed logic diagram of isomorphism

approximation engine, the designer can change the number of samples and the maximum iteration per sample, whereas in coherence vector engine, the designer sets the temperature, relaxation time, total simulation time, etc. [18]. In the proposed QCA implementation, the coherence vector model is chosen. In addition, we selected the Euler transformation, randomized simulator and GAUSS type clock signal. The values of the simulation setups parameters are listed in Table 1. Area, power, delay of various functional modules in AES SBOX are evaluated. The results and discussion are described in two parts; the first part describes the verification of functional operation of GF arithmetic with help of the waveform and the second part analyzes the cell count, energy dissipation, area and latency model GF functions in QCA.

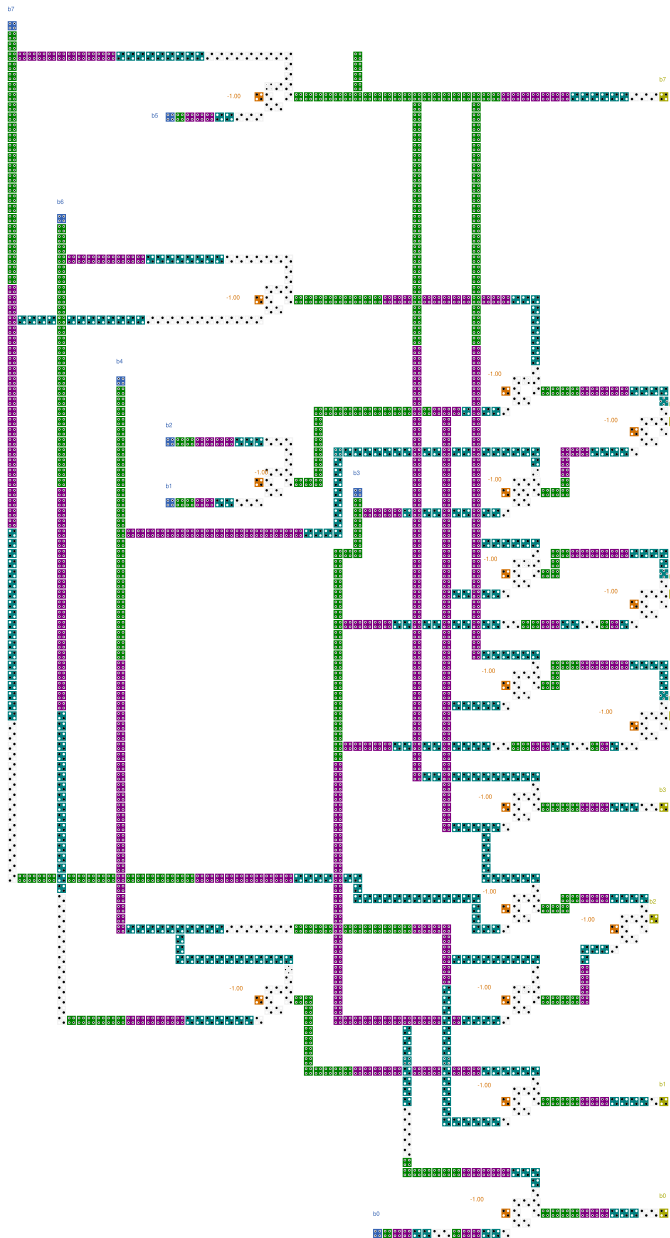


Fig. 20 QCA implementation of Isomorphism

5.2 Simulation Results

GF Squarer simulation result is shown in Fig. 21. In simulation, inputs are denoted in b0, b1, b2, b3 and outputs are mentioned in k0, k1, k2 and k3. Here outputs are

Table 1 Simulation setup parameters

Description/parameters	Value
Temperature	1.000K
Relaxation time	1.000e-015 s
Clock period	4.00e-012s
Input period	4.00e-012s
Total simulation time	5.00e-11s
Clock high	9.800e-22 J
Clock low	3.8000e-023 J
Clock shift	0
Clock slope	1.000e-012s
Radius effect	80.00nm
Relative permittivity	12.900
Layer separation	11.500mm

obtained after one clock period because of latency of the GF squarer being one clock cycle. The result is verified based on Eq. 5.

A two-bit GF multiplication uses two operand w and b . The output is denoted in 4 bits. It took 4 clock cycle to converge the results. The output is valid after the 4 clock signal (Figs. 22 and 23).

A simplest GF lambda multiplication uses three XOR gates. The three input XOR gate is implemented by using two input XOR gates. Hence, the delay has increased and it took 4 cycle to get the valid output. The valid outputs are shown in 23.

In Fig. 24, GF (ϕ) multiplication output waveform is shown. For example, the input '1' is circled and the same will be available after one clock signal. The output is circled for verification.

To verify the isomorphism function, Figs. 25 and 26 are viewed separately. A valid 8-bit outputs are available after 3 clock cycle.

5.3 Area usage, Latency, Energy Dissipation, & Quantum Cost

The proposed QCA-based GF function module is analyzed in terms of parameters such as complexity, total area, cell area usage, energy dissipation, average energy/cycle, no. of clock cycles and quantum cost. The above said parameters are grouped into four major categories, namely area, energy, clock operation and quantum cost [33]. The values of these performance parameters obtained for the QCA-based GF functional modules are listed in Table 2.

Design complexity is given by the number cells used in the design. Total complexity of the proposed design is 1554. The percentage distribution of the design complexity of the GF functions is shown in Fig. 27. It can be observed that isomorphic function has the highest design complexity.

Total area in the design is calculated as the rectangular area occupied by the QCA cells. Figure 28 shows the percentage distribution of total area of GF functions

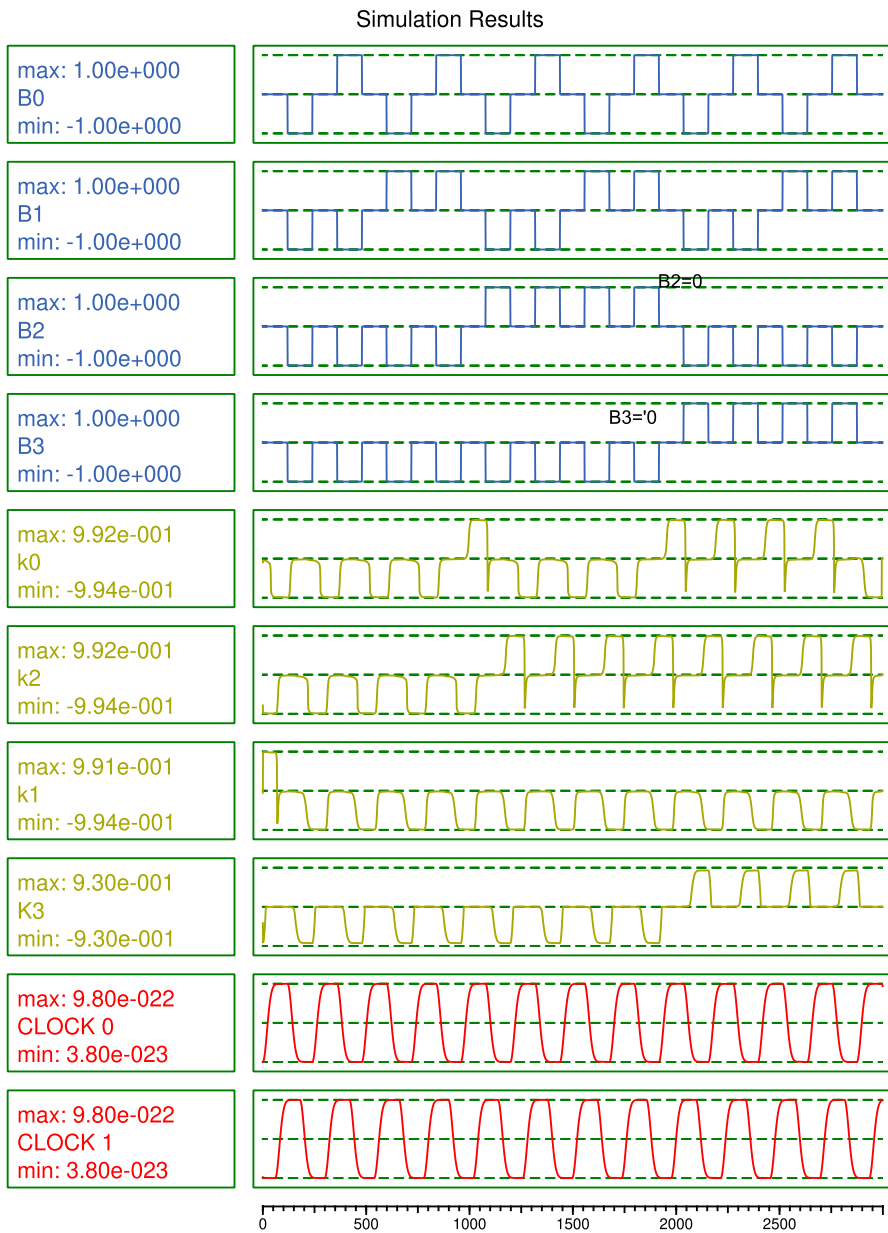


Fig. 21 Simulation waveform of GF Squarer

and as expected isomorphic function has the maximum total area due to its highest design complexity

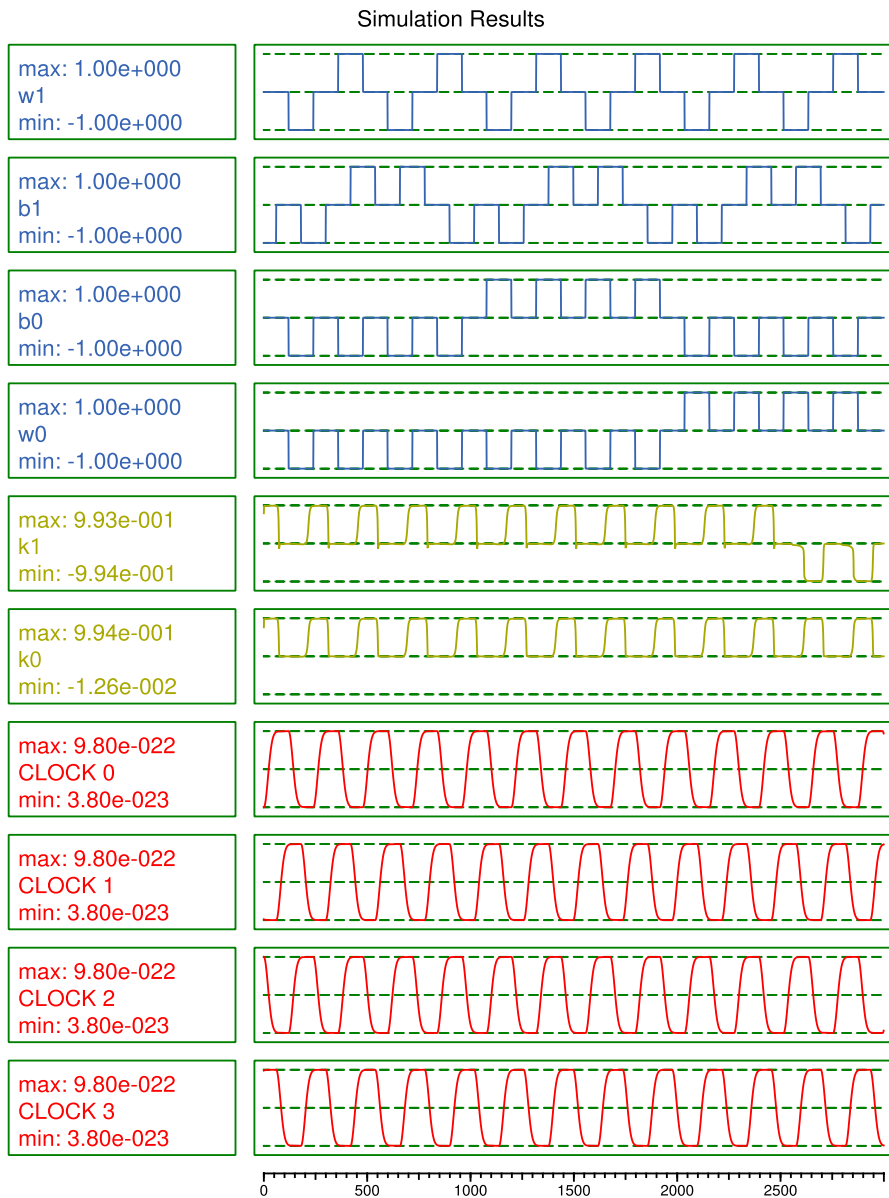


Fig. 22 Simulation waveform of GF multiplication

The cell area of the design is product of design complexity and one cell area (ie. $18\text{nm} \times 18\text{nm} = 324 \text{ nm}^2$). Total cell area of the proposed model is 0.614006 nm^2 . Area usage of proposed design model has a maximum of 23 % and minimum of 12.9 %.

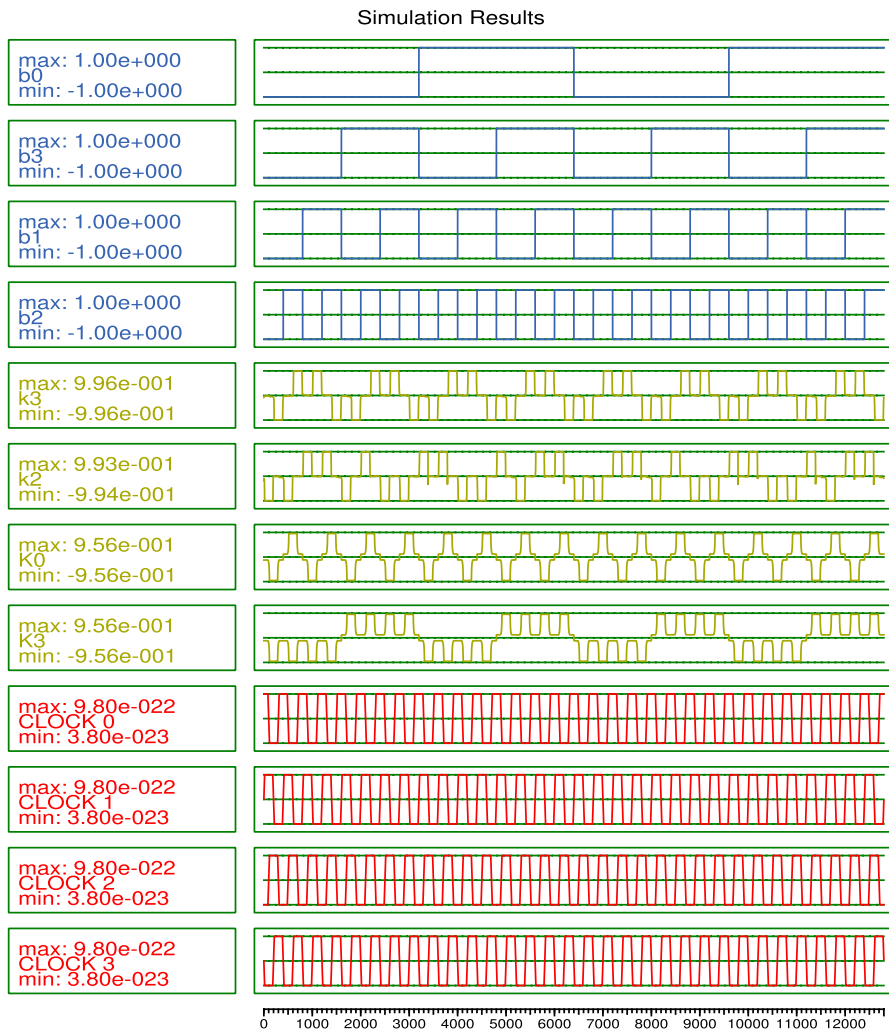


Fig. 23 Simulation waveform of GF Lambda λ Multiplication

5.3.1 Calculation of cost functions

In any system design, cost plays a significant parameter to choose the design. Here, in QCA too, [33], we consider the area delay cost function, which is calculated by the product of area and delay or latency of the design, i.e.,

$$\text{Area - delay cost} = \text{Area} * \text{Delay} \quad (10)$$

The area is specified in nm^2 and delay or latency in clock cycle. The obtained values are given in Table 3.

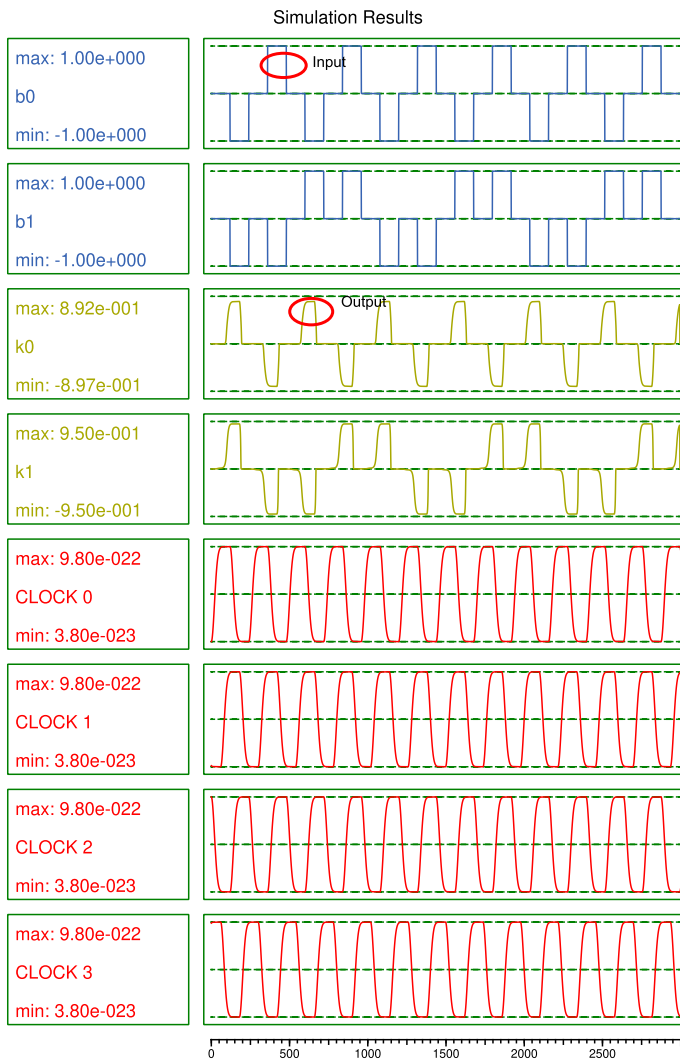


Fig. 24 Simulation waveform of GF(ϕ) Multiplication

5.3.2 Energy Dissipation

To analyze the energy dissipation of proposed functions, the energy calculation of single cell QCA is calculated. In each cell in a QCA circuit, energy is considered an energy bath [36]. While circuit is working in a given clock cycle, cell-to-cell energy transfer occurs. As a result, the energy transformations for each QCA energy ‘bath’ are to be calculated using the tool QCADesigner-E [10]. A detail energy dissipation and average energy per cycle are presented in Table 2. Latency is defined by the delay incurred in transferring the data from input to output. For example, refer to

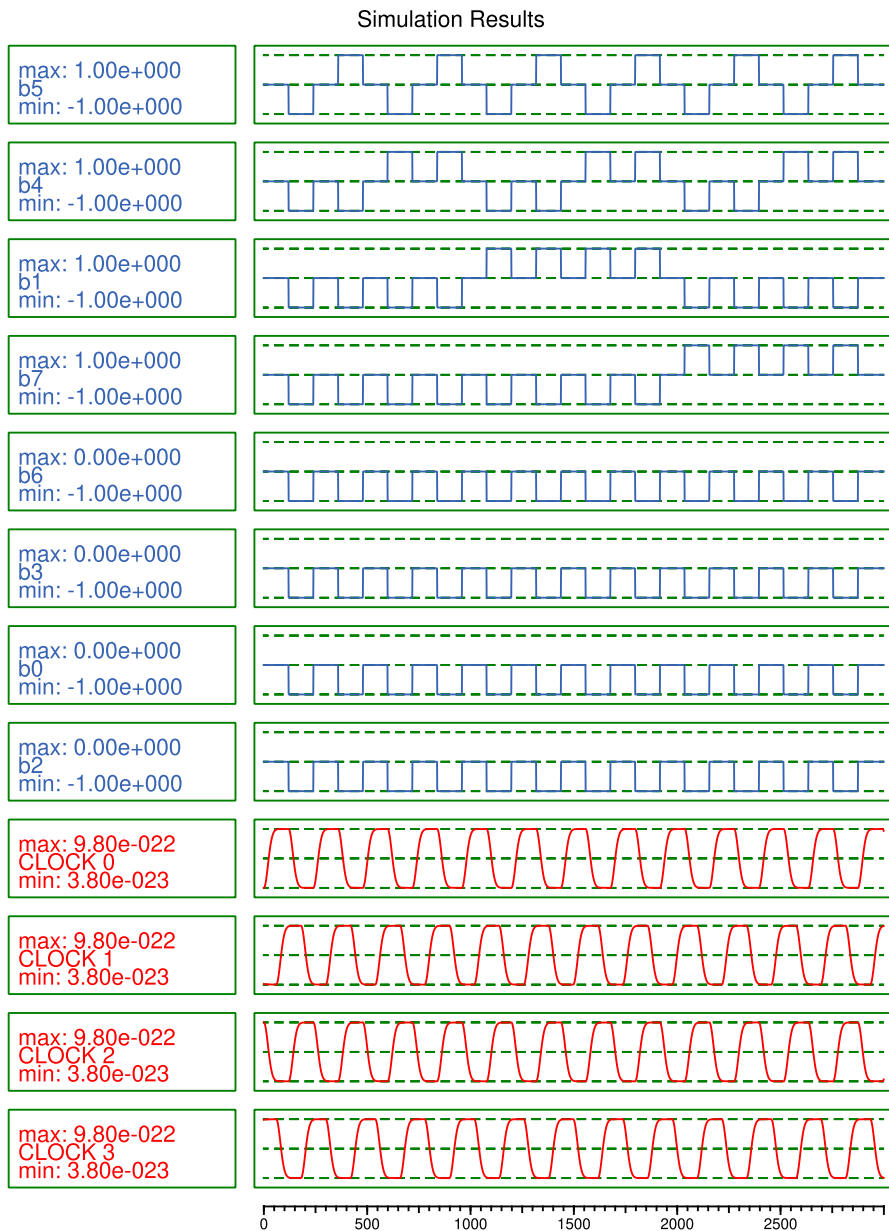


Fig. 25 Simulation waveform of isomorphism—input

Fig. 11. In our proposed GF square, k_0 and k_1 Figure 11 has 1 clock cycle in critical path, whereas k_2 and k_3 have 0.75 and 0.5 clock cycle. So, the critical path delay of the circuit has maximum delay of 1 clock cycle. Likewise, latency of other functional models is calculated and tabulated in Table 2.

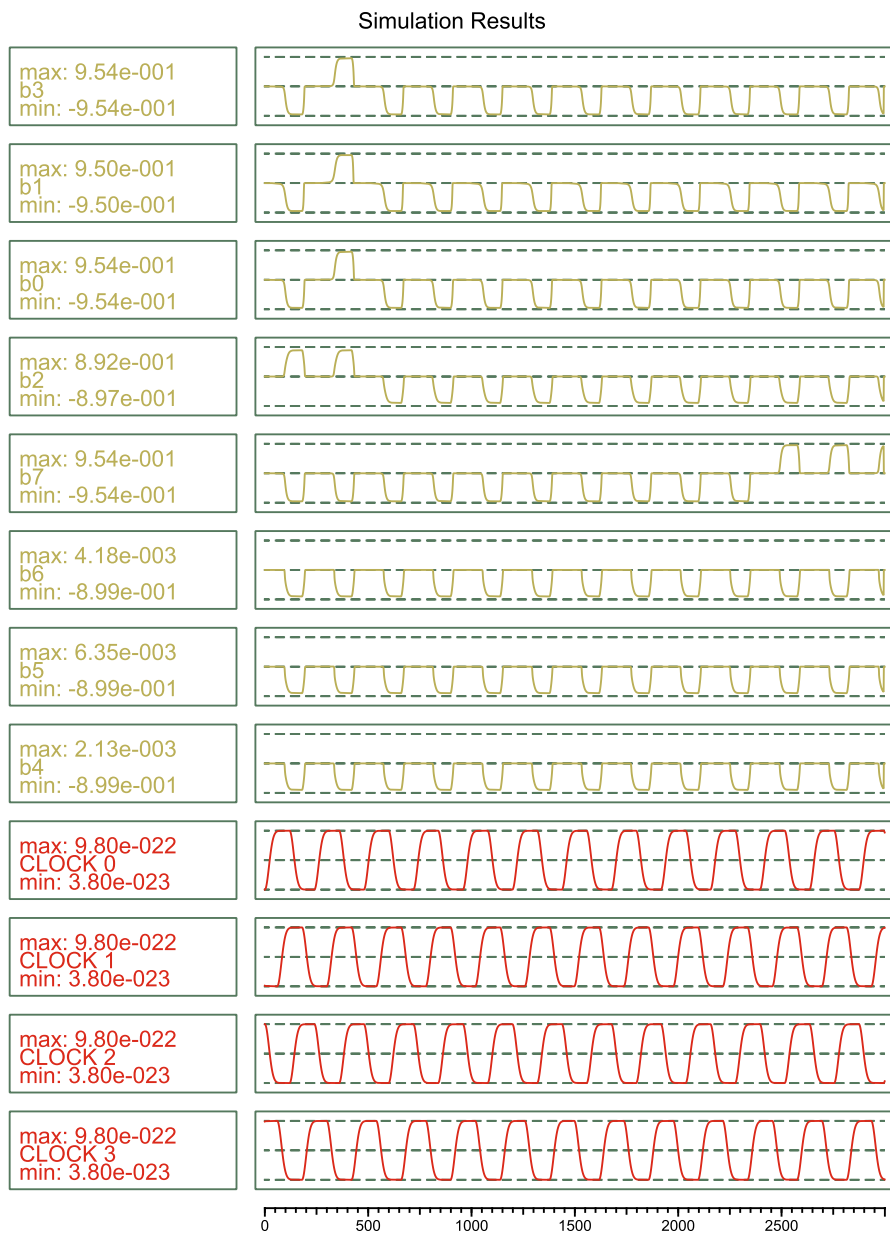


Fig. 26 Simulation waveform of isomorphism—output

5.3.3 Energy dissipation of GF functions

Energy dissipation of GF functions (multiplication, pi multiplication, square, lambda and isomorphism) is presented in Tables 4, 5, 6, 7 and 8. Here E_bath_total

Table 2 Energy dissipation, area, latency & quantum cost of QCA-based GF functions modules

Design/ function module	GF squarer	GF multiplication	GF λ multiplication	GF ϕ multiplication	Isomorphic function
No. of input bits	4	4	4	2	8
No. of output bits	4	2	4	2	8
No. of XOR gate	3	3	2	1	17
No. of AND gate	0	5	0	0	0
No. of QCA cell	63	168	59	27	1340
Total area (mm^2)	120	280	97.58	38	3377
Cell area (nm^2) 324	0.02041	0.0528	0.019116	0.008748	0.4342
Area usage %	17.01	18.9	19.6	23.01	12.9
Energy dissipation (10-3eV)	0.454	0.394	0.415	0.566	347
Average energy per cycle (10-3eV)	3.58	5.58	3.77	1.51	0.316
Clocking Zone	3	4	4	2	4
Delay/ latency (clock cycle)	1	1.75	1.5	0.5	3
Quantum cost (total area $\times$ latency)	0.12	0.49	0.14637	0.019004	10.131

Fig. 27 Percentage distribution of design complexity of GF functions

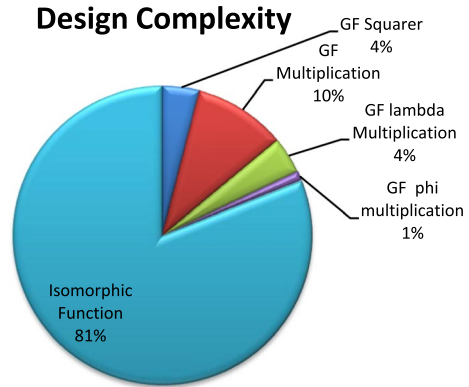


Fig. 28 Percentage distribution of area usage of GF functions

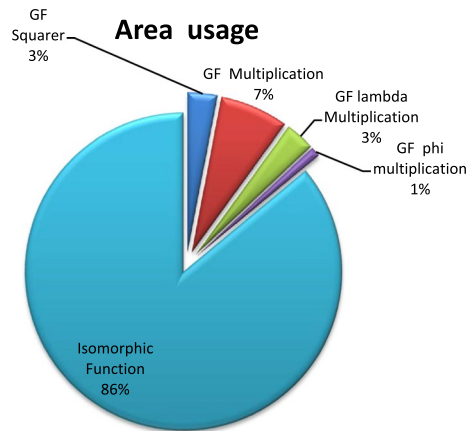


Table 3 Area–delay cost of QCA-based functioning modules

Design	GF squarer	GF multi- plication	GF λ multiplication	GF ϕ multiplication	Isomor- phic function
Area	120	280	97.58	38	3377
Delay	1	1.75	1.5	0.5	3
Area delay cost (nm^2/sec)	120	490	146.37	19.0035	10131

is the total energy transfer in each QCA cell for each clock cycle. $E_{\text{clk_total}}$ is the between inter-cell transfer in the presence of clock signal. $E_{\text{error_total}}$ is the difference between $E_{\text{bath_total}}$ and $E_{\text{clk_total}}$. Sum_Ebath represents the sum of total energy transferred for the entire simulation [10]. To complete the entire simulation, GF multiplication, GF π multiplication and isomorphism took 6 iterations, and GF Square and GF_{λ} took 8 iterations. The simulation is done using coherence vector model with 500000 samples (Fig. 29).

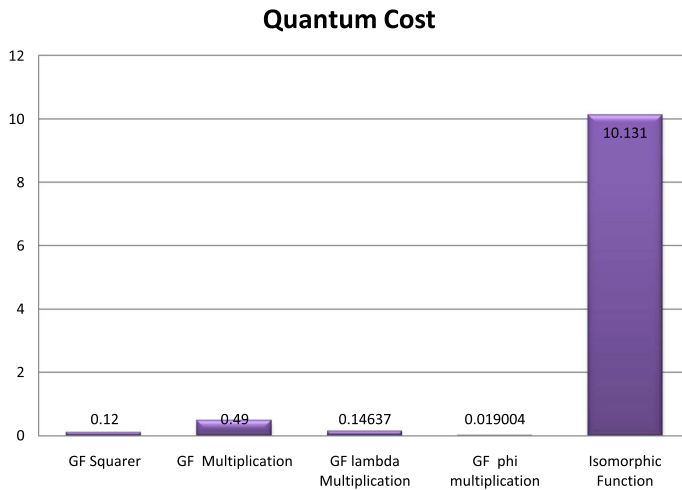


Fig. 29 Quantum cost analysis of GF functions

Table 4 Energy dissipation in GF multiplications

E_bath_total (eV)	E_clk_total (eV)	E_Error_total (eV)	Sum_Ebath (eV)	Avg_Ebath (eV)
$4.26 * 10^{-3}$	$1.19 * 10^{-3}$	$-3.43 * 10^{-4}$	$6.14 * 10^{-2}$ Error: $-5.40 * 10^{-3}$	$5.58e * 10^{-3}$ (Error: $-4.91 * 10^{-4}$)
$5.64 * 10^{-3}$	$-4.69 * 10^{-4}$	$-4.97 * 10^{-4}$		
$5.47 * 10^{-3}$	$-2.41 * 10^{-4}$	$-4.78 * 10^{-4}$		
$5.63 * 10^{-3}$	$-3.03 * 10^{-4}$	$-4.96 * 10^{-4}$		
$5.07 * 10^{-3}$	$+2.19 * 10^{-4}$	$-4.35 * 10^{-4}$		
$6.16 * 10^{-3}$	$-1.07 * 10^{-3}$	$-5.56 * 10^{-4}$		
$6.29 * 10^{-3}$	$-1.21 * 10^{-3}$	$-5.70 * 10^{-4}$		
$6.36 * 10^{-3}$	$-8.68 * 10^{-4}$	$-5.82 * 10^{-4}$		
$5.10 * 10^{-3}$	$-1.86 * 10^{-5}$	$-4.37 * 10^{-4}$		
$5.49 * 10^{-3}$	$-3.17 * 10^{-4}$	$-4.81 * 10^{-4}$		
$5.90 * 10^{-3}$	$-7.62 * 10^{-4}$	$-5.26 * 10^{-4}$		

From Table 4, 5, 6, 7, 8, it can be observed that average Ebath error is very low for GF ϕ multiplication (Error : $-1.56 * 10^{-4}$) and highest value for GF multiplication (Error : $-4.91 * 10^{-4}$). An error value is increased because the latency of the GF multiplication is large compared to GF ϕ multiplication. The latency of these circuits is 4 and 2, respectively. Similarly, it can be observed that the sum of Ebath value also be increased due to this latency. In another observation, the design complexity and quantum cost also be increased.

Table 5 Energy dissipation in GF pi multiplications

E_bath_total (eV)	E_clk_total (eV)	E_Error_total (eV)	Sum_Ebath (eV)	Avg_Ebath (eV)
$1.72 * 10^{-3}$	$2.15 * 10^{-4}$	$-1.77 * 10^{-4}$	$1.66 * 10^{-2} \text{eV} (\text{Error} : -1.71 * 10^{-3})$	$1.51 * 10^{-3}$ (Error: $-1.56 * 10^{-4}$)
$1.38 * 10^{-3}$	$9.57 * 10^{-4}$	$-1.43 * 10^{-4}$		
$1.12 * 10^{-3}$	$7.98 * 10^{-4}$	$-1.13 * 10^{-4}$		
$1.96 * 10^{-3}$	$1.37 * 10^{-4}$	$-2.06 * 10^{-4}$		
$1.72 * 10^{-3}$	$2.15 * 10^{-4}$	$-1.77 * 10^{-4}$		
$1.38 * 10^{-3}$	$9.57 * 10^{-4}$	$-1.43 * 10^{-4}$		
$1.12 * 10^{-3}$	$7.98 * 10^{-4}$	$-1.13 * 10^{-4}$		
$1.96 * 10^{-3}$	$1.37 * 10^{-4}$	$-2.06 * 10^{-4}$		
$1.72 * 10^{-3}$	$2.15 * 10^{-4}$	$-1.77 * 10^{-4}$		
$1.38 * 10^{-3}$	$9.57 * 10^{-4}$	$-1.43 * 10^{-4}$		
$1.12 * 10^{-3}$	$7.98 * 10^{-4}$	$-1.13 * 10^{-4}$		

Table 6 Energy dissipation in GF squarer

E_bath_total (eV)	E_clk_total (eV)	E_Error_total (eV)	Sum_Ebath (eV)	Avg_Ebath (eV)
$3.78 * 10^{-3}$	$6.98 * 10^{-4}$	$-5.37 * 10^{-4}$	$3.94e - 002 (\text{Error} : -4.08 * 10^{-3})$	$3.58 * 10^{-3}$ (Error: $-3.70 * 10^{-4}$)
$4.13 * 10^{-3}$	$3.24 * 10^{-4}$	$-4.17 * 10^{-4}$		
$4.64 * 10^{-3}$	$-1.54 * 10^{-3}$	$-4.72 * 10^{-4}$		
$4.59 * 10^{-3}$	$1.64E - 05$	$-4.68 * 10^{-4}$		
$2.96 * 10^{-3}$	$1.74 * 10^{-3}$	$-2.86 * 10^{-4}$		
$3.31 * 10^{-3}$	$1.39 * 10^{-3}$	$-3.28 * 10^{-4}$		
$3.82 * 10^{-3}$	$-4.71 * 10^{-4}$	$-3.83 * 10^{-4}$		
$3.94 * 10^{-3}$	$1.09 * 10^{-3}$	$-3.97 * 10^{-4}$		
$2.34 * 10^{-3}$	$2.34 * 10^{-3}$	$-2.16 * 10^{-4}$		
$2.69 * 10^{-3}$	$1.99 * 10^{-3}$	$-2.58 * 10^{-4}$		
$3.20 * 10^{-3}$	$1.40 * 10^{-4}$	$-3.14 * 10^{-4}$		

6 Conclusion and Future Works

AES encryption is a notable cryptography algorithm in security. Energy estimation is essential for QCA circuits to analyze their performance. This paper provides energy estimation as well as design complexity with success. We analyzed the impact of energy and area of the GF arithmetic functions for AES SBOX. In future work, we intend to investigate combining all GF function modules by using multilayer and cross over configuration for 8-bit SBOX operation for advanced

Table 7 Energy dissipation in GF lambda multiplications

E_bath_total (eV)	E_clk_total (eV)	E_Error_total (eV)	Sum_Ebath (eV)	Avg_Ebath (eV)
$5.67 * 10^{-3}$	$-2.33 * 10^{-3}$	$-5.88 * 10^{-4}$	$4.15 * 10^{-2}$ (Error: $-4.19 * 10^{-3} eV$)	$3.77 * 10^{-3}$ (Error: $-3.81 * 10^{-4}$)
$3.76 * 10^{-3}$	$2.87 * 10^{-4}$	$-3.79 * 10^{-4}$		
$3.42 * 10^{-3}$	$1.73 * 10^{-4}$	$-3.41 * 10^{-4}$		
$6.02 * 10^{-3}$	$-2.36 * 10^{-3}$	$-6.28 * 10^{-4}$		
$3.50 * 10^{-3}$	$3.02 * 10^{-4}$	$-3.48 * 10^{-4}$		
$3.14 * 10^{-3}$	$1.11 * 10^{-3}$	$-3.11 * 10^{-4}$		
$2.80 * 10^{-3}$	$9.97 * 10^{-4}$	$-2.74 * 10^{-4}$		
$4.09 * 10^{-3}$	$2.88 * 10^{-4}$	$-4.19 * 10^{-4}$		
$3.39 * 10^{-3}$	$2.02 * 10^{-4}$	$-3.37 * 10^{-4}$		
$3.04 * 10^{-3}$	$1.01 * 10^{-3}$	$-3.00 * 10^{-4}$		
$2.70 * 10^{-3}$	$8.96 * 10^{-4}$	$-2.62 * 10^{-4}$		

Table 8 Energy dissipation GF isomorphism

E_bath_total (eV)	E_clk_total (eV)	E_Error_total (eV)	Sum_Ebath (eV)	Avg_Ebath (eV)
$3.12 * 10^{-2}$	$-2.95 * 10^{-2}$	$-2.39 * 10^{-3}$	$3.47 * 10^{-1}$ (Error: $-2.64 * 10^{-2}$)	$3.16 * 10^{-2}$ (Error: $-2.40 * 10^{-3}$)
$3.21 * 10^{-2}$	$-3.09 * 10^{-2}$	$-2.43 * 10^{-3}$		
$3.10 * 10^{-2}$	$-2.95 * 10^{-2}$	$-2.34 * 10^{-3}$		
$3.15 * 10^{-2}$	$-3.00 * 10^{-2}$	$-2.38 * 10^{-3}$		
$3.15 * 10^{-2}$	$-3.02 * 10^{-2}$	$-2.38 * 10^{-3}$		
$3.15 * 10^{-2}$	$-3.02 * 10^{-2}$	$-2.39 * 10^{-3}$		
$3.13 * 10^{-2}$	$-3.00 * 10^{-2}$	$-2.37 * 10^{-3}$		
$3.15 * 10^{-2}$	$-3.00 * 10^{-2}$	$-2.39 * 10^{-3}$		
$3.15 * 10^{-2}$	$-3.03 * 10^{-2}$	$-2.39 * 10^{-3}$		
$3.20 * 10^{-2}$	$-3.05 * 10^{-2}$	$-2.45 * 10^{-3}$		
$3.23 * 10^{-2}$	$-3.10 * 10^{-2}$	$-2.49 * 10^{-3}$		

development secure communication. In addition, to improve the energy consumption, it will be implemented in Launder's clock zone and Bennet's clock zone and also possible to compare performance in throughput.

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Declarations

Conflict of Interest All authors have no conflict of interest.

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